



A thought-to-text communication system that “speaks” the thoughts

Stanford University researchers have published a paper claiming a record in thought-to-text communication through a brain-computer interface (BCI), with a subject being able to “talk” at a rate of 62 words per minute. Their work consists of “neuroprosthesis”—a brain-computer interface (BCI) that uses intracortical microelectrode arrays to capture high-resolution recordings of its users’ brain activity associated with speech. Researchers tested their work on an individual whose illness resulted in an inability to generate intelligible speech. The device enabled the subject to think words and have them decoded at a rate of 62 words per minute, which is over three times faster than previous state-of-the-art BCI speech systems. The device accuracy was measured 9.1% for a limited 50-word vocabulary and increased to 23.8% for 125,000 word-vocabulary, which makes it usable.

New system-on-chip identifies and manages neurological disorders

Researchers at Ecole Polytechnique Fédérale de Lausanne have developed a system called NeuralTree to identify and manage neurological disorders with its machine learning ability. NeuralTree is a closed-loop neuromodulation system-on-chip that can detect and alleviate disease symptoms. The system can extract and classify a broad set of biomarkers from real patient data and animal models of disease in-vivo, leading to a high degree of accuracy in symptom prediction. The system consists of a 256-channel high-resolution input sensing array and an energy-efficient machine learning processor. It can detect a broader range of symptoms than other devices, which until now have focused primarily on epileptic seizure detection. The chip’s machine learning algorithm was trained on datasets from both epilepsy and Parkinson’s disease patients, and accurately classified pre-recorded neural signals from both categories.



NeuralTree (Credit: Alain Herzog)

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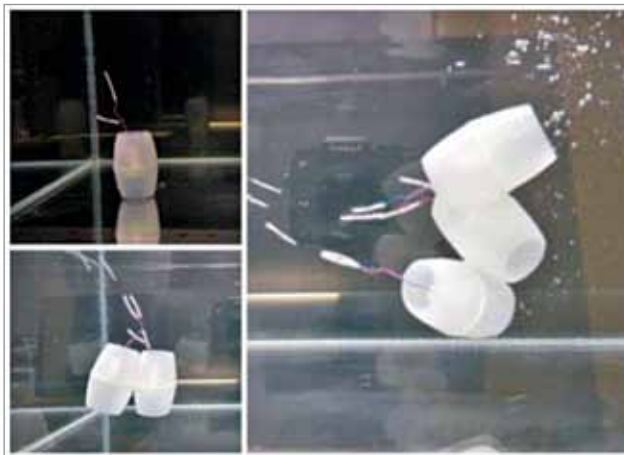
Detecting radio signals from distant galaxies made possible

Astronomers from McGill University in Canada and the Indian Institute of Science (IISc) in Bengaluru have used

data from the Giant Metrewave Radio Telescope (GMRT) in Pune to detect a radio signal from atomic hydrogen in a distant galaxy at redshift $z = 1.29$. This is the first confirmed detection of strong lensing of 21cm emission from a galaxy. This detection was made possible by a phenomenon called gravitational lensing, in which the light emitted by the source is bent due to the presence of another massive body, such as an early type elliptical galaxy, between the target galaxy and the observer, effectively resulting in the “magnification” of the signal. The atomic hydrogen mass of this particular galaxy is almost twice as high as its stellar mass. Results demonstrate the feasibility of observing atomic gas from galaxies at cosmological distances in similar lensed systems with a modest amount of observing time.

Underwater robots engineered to operate in extra-terrestrial oceans

Scientists at the University of Bristol have drawn on the design and life of a mysterious zooplankton to develop underwater robots. These robotic units called RoboSalps, after their animal namesakes, have been engineered to operate in unknown and extreme environments like extra-terrestrial oceans. RoboSalps have light, tubular bodies and can link to each other to form “colonies” which gives them



RoboSalps in action (Credit: Valentina Lo Gatto)

new capabilities that can only be achieved because they work together. These simple modules can be combined into ‘colonies’ that are much more robust and have the potential to carry out complex tasks. When swimming on their own, RoboSalps modules are difficult to control, but after joining them together to form colonies, they become more stable and show sophisticated movements. By joining them together, scientists can obtain a redundant system that is robust against any possible failure.